TUSHANG PATEL

Embedded Firmware Engineer

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SUMMARY

Embedded Firmware Engineer with two years of hands-on experience building real-time, production-grade firmware for medical, automotive, and IoT devices. Skilled in C/C++, FreeRTOS, and multi-protocol communication (BLE, GSM, GPS, MQTT, Modbus). Proven track record across the full firmware lifecycle: bootloaders, OTA updates, sensor integration, and hardware-firmware bring-up. Comfortable owning a product from schematic review through field-ready deployment in fast-paced startup environments.

TECHNICAL SKILLS

- Languages: C, C++, Python, MicroPython, HTML, CSS
- Embedded Systems: FreeRTOS, Multithreading, Memory Management, ISR, DMA, Custom Bootloader Development, Firmware Partitioning, Secure Boot Flow
- MCUs & Boards: STM32, ESP32, PY32, Arduino Uno, Arduino Mega
- Protocols: BLE, Wi-Fi, GSM, GPS, AGPS, MQTT, HTTP, WebSocket, OTA, Modbus RTU
- Interfaces & Peripherals: ADC, DAC, GPIO, I^2C, SPI, UART, I^2S, PWM, Timers
- Hardware Modules: RFID, Cellular Modems (MC60, EC200)
- Tools & IDEs: STM32CubeIDE, STM32CubeMX, STM32CubeProgrammer, ESP-IDF, Keil µVision, QuecPython IDE, nRF Connect SDK, VS Code
- Debugging & Hardware: Schematic reading, logic analyzers, oscilloscopes, hardware-firmware bring-up
- Version Control: Git, GitHub, GitLab

WORK EXPERIENCE

Embedded Firmware Engineer | OOB Services, Ahmedabad, Gujarat **Jan 2025 – Present**

- Design and implement embedded applications in C, C++, Python, and MicroPython, optimizing firmware for constrained, hardware-oriented targets.
- Build FreeRTOS-based real-time systems with task scheduling, queues, semaphores, and dynamic memory management for reliable multitasking.
- Develop multithreaded MicroPython applications for concurrent sensor processing and communication, enabling simultaneous real-time data handling.
- Deliver firmware across a diverse hardware portfolio — STM32, ESP32, PY32, Arduino Uno, and Arduino Mega — adapting design to each platform's constraints.
- Lead firmware debugging, board bring-up, and hardware validation, working directly with hardware teams to resolve integration issues before production.

Embedded Firmware Engineer | Electrogenics Innovations, Ahmedabad, Gujarat **Jan 2024 – Dec 2024**

- Developed C/C++ firmware for STM32-based neonatal medical devices, prioritizing reliability and patient safety in a regulated device environment.
- Implemented RTOS-based multitasking and peripheral drivers (UART, SPI, I^2C) to support deterministic, real-time device behavior.
- Integrated biomedical sensors (NIBP, SpO_2) and HMI displays for continuous real-time patient monitoring.
- Built a custom Arduino IDE library for W25Q16BV SPI Flash, reused across multiple internal projects to speed up flash integration.
- Contributed core firmware modules to the Bubble CPAP medical device program from design through hardware validation.

KEY PROJECTS

Web-Based Train Audio Announcement System — OTA & Custom Bootloader

- Built a train announcement system letting operators upload MP3 files and configure routes through a web interface, with SD-card audio storage and route/signal-triggered playback.
- Designed a custom bootloader with SoftAP-based OTA, enabling secure over-the-air firmware updates via startup button logic — no physical reflashing required in the field.

IoT-Based Smart Bus Ticketing & Tracking System — Cellular MQTT

- Developed an automated ticketing and GPS tracking system on the EC200U cellular module with real-time MQTT communication for ticket data and location updates.
- Engineered for continuous connectivity in mobile, signal-variable environments typical of public transit.

Embedded High-Voltage Power Distribution & Safety Control System — EV Platform

- Built an automotive-grade control system managing battery, inverter, DC/AC charging, and auxiliary loads via I^2C GPIO expanders.
- Implemented fault detection, protection interlocks, and JSON-based diagnostics to support safe high-voltage operation.

BLE-Enabled Smart Cycle Control & Location Tracking System

- Designed a BLE-controlled smart cycle system supporting Start, Hold, and End ride states with autonomous state handling and secure command processing.
- Integrated a Quectel MC60 modem for GPS tracking and MQTT-based data upload.

Pulse Oximeter & Vital Sign Monitoring System

- Designed an embedded pulse oximeter for real-time SpO_2 and heart-rate monitoring, interfacing biomedical sensors via embedded C/C++.
- Implemented signal-processing and data-filtering algorithms to improve measurement accuracy, with firmware built around medical-device reliability standards.

DMA-Driven I^2S & Timer-PWM Audio Playback (W25 SPI Flash)

- Programmed WAV audio into external W25 SPI Flash and implemented DMA circular-mode I^2S playback for continuous, low-latency, glitch-free streaming.
- Built an alternate 44 kHz timer-based PWM playback path for DAC-less audio output, with optimized SPI read timing.

Additional Projects: Programmable Educational Robot with motion sequencing & drawing mode · Human-Body Controlled DAC-Based LED System with thermal fail-safe · RFID-Based Multi-Door Smart Security System · Modbus RTU Master–Slave Communication (STM32H533RE)

EDUCATION

Bachelor of Engineering — Electronics & Communication Engineering April 2024
Government Engineering College, Gandhinagar | CGPA: 8.16